Homework 11: Inference for Regression in R

Math 141, Week 12

Your name here

Assigned: Thursday, November 19th

Due: TUESDAY, December 2nd at 9:00am on Gradescope as a PDF

In this homework assignment you will:

- Do inference for regression with both simulation-based and theory-based methods

Add any collaborators here!

I collaborated with…

Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this class. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

```r
#load the necessary packages
library(tidyverse)
library(knitr)
library(moderndive)
library(infer)
library(openintro)
```

Exercises

Exercise 1: Metacognition

We've covered a lot of ground this semester! Here's a brief summary of topics, grouped into 5 units:

- Unit 1: Data visualization, summary statistics, probability, sampling methods, study design
- Unit 2: Linear regression (simple and multiple), quantitative and categorical explanatory variables, interaction effects
- Unit 3: Sampling distributions, bootstrap distributions, confidence intervals, hypothesis testing
- Unit 4: Random variables, the Central Limit Theorem, theory-based inference for proportions, means, difference in proportions, difference in means
- Unit 5: Inference for regression with theory and bootstrapping

a. Pick 3 distinct "pairs" of topics from *different* units. For each pair, describe in 1-2 complete sentences how the concepts in those units are related. (ex. Data visualization and linear regression are related, because we should always use data visualization techniques to assess LINE assumptions. Residual plots and y-vs-x scatterplots help us assess linearity and equal variance, and histograms or Q-Q plots help us assess normality.)

b. Which of the 3 topics did you find most challenging? For each topic, identify one related homework or lab exercise that you want to revisit before the final exam.

Exercise 2: Incorporating Gender

For the remaining exercises in this assignment, we'll build on our exploration of the professor evaluation data from Lab 11. Run the chunk below to load the data set again.

```r
data(evals)
```

In lab, we assessed a simple linear regression that used the beauty score to predict the instructor evaluation score. However, it could be the case that beauty has different effects on evaluations depending on the professor's gender. In order to see if beauty is a meaningful predictor of professor score after we've accounted for the gender of the professor, we should build a model that includes both beauty and gender terms.

a. Use `lm()` to create a regression model predicting `score` as a function of `bty_avg` and `gender`. Display the estimated coefficients using the `get_regression_table()` function. [Code only for this question]

b. Write down the equation for the linear model, being sure to include the estimated coefficients in the equation. [Text only for this question]

c. Interpret the (i) intercept and the (ii) slope on `bty_avg` *in context*. [Text only for this question]

d. Now that we're controlling for `gender`, have the results of the hypothesis test for whether `bty_avg` is a meaningful predictor of `score` changed? **How should we interpret this?** [Text only for this question]

e. Using the regression table from Exercise 2 part (a), answer the following:

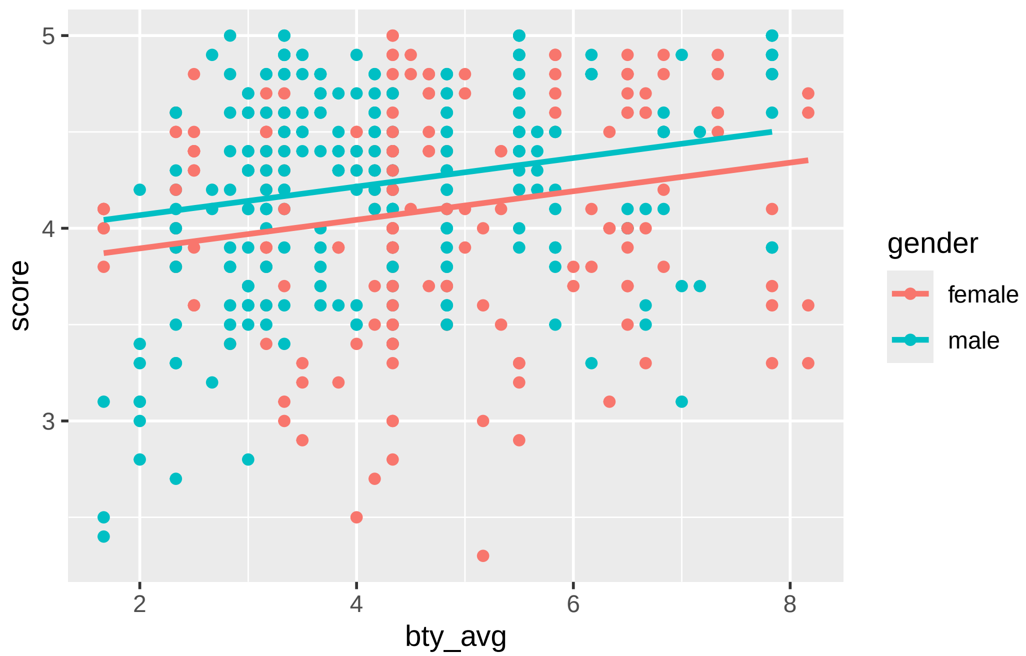
(i) What is the equation of the line corresponding to male professors?

(ii) What is the equation of the line corresponding to female professors?

(iii) For two professors who received the same beauty rating, which gender tends to have the higher course evaluation score?

The following code chunk plots the regression lines from the multiple regression model corresponding to male and to female professors. You don't need to include this in your response, but the graph may provide intuition for the answers you report. [Text only for this question]

```
ggplot(evals, aes(x = bty_avg, y = score, color = gender)) +
  geom_point() +
  geom_parallel_slopes(se = FALSE)
```



Exercise 3: Incorporating Age

a. Create a new model that includes `age`, `bty_avg`, and `gender`, to predict `score`, and display the regression table. [Code only for this question]

b. The slope on `age` in the model constructed in Exercise 3(a) is a relatively small number. Does this mean that `age` is not a useful predictor for this model? Explain, appealing to both formal inference and the context. [Text only for this question]

c. Using plots and contextual knowledge, assess if the model constructed in Exercise 3(a) satisfies **each** of the LINE conditions. [Code and text for this question]

d. Based on your checks for the LINE assumptions, should we rely on theory-based inference, or take a simulation-based approach? Explain. [Text only for this question]

Exercise 4: Simulation-based Inference for Regression

a. Regardless of what you found in Exercise 3, we will now use bootstrapping to conduct inference on parameters of the model we constructed there. Construct a 80% confidence interval for the coefficient for `bty_avg` using the bootstrap with `reps = 1000` and the percentile method (this may take a minute to run). **Be sure to state the confidence interval, and interpret the confidence interval in the context of what your statistic represents.** [Code and text for this question]

(*Hint: An example of code for simulation-based inference for regression can be found in the slides from 4/15. When adapting this code, make sure you are grabbing the estimate for* `bty_avg` *- check the order of your coefficients in Exercise 3a.*)

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b. Conduct a hypothesis test of the coefficient for `bty_avg` being equal or not equal to 0 using the bootstrap distribution you just created. Use a significance level of $\alpha = 0.01$. **Be sure to state your hypotheses, calculate a p-value, and interpret the p-value in context.** [Code and text for this question]

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c. Do you think it would be reasonable to generalize these conclusions to a larger population? Would it be reasonable to draw causal conclusions about how beauty and student evaluation scores are related? Explain. [Text only for this question]
